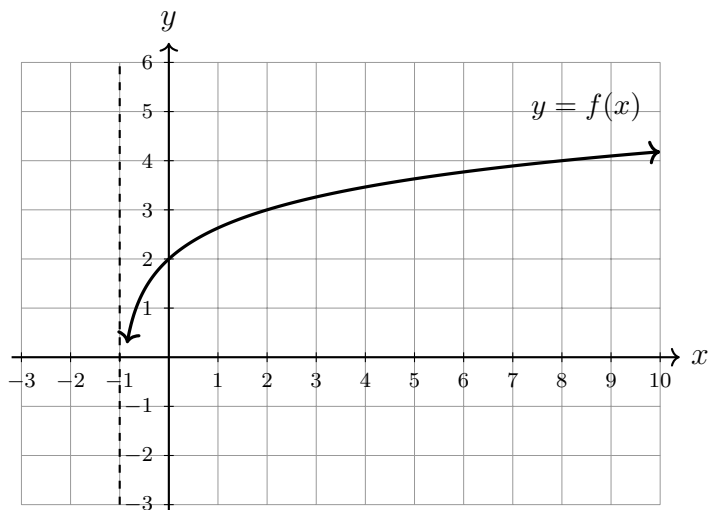


7.10 PreQuiz: Logarithmic and Exponential Functions (calculator)

1. [Maximum mark: 6]

The graph of $f(x) = \log_3(x + 1) + 2$ is shown below.



- Write down the equation of the vertical asymptote of f . [1]
- Find the x -intercept of the graph of f . [3]
- Write down the range of f . [1]
- State the coordinates of the point on the graph of f whose x -coordinate is 8. [1]

.....

.....

.....

.....

.....

.....

2. [Maximum mark: 6]

Let $g(x) = 4 \cdot 2^{x-1} - 3$.

(a) Write down the equation of the horizontal asymptote of $y = g(x)$. [1]

(b) Find $g(3)$. [2]

(c) Find an expression for $g^{-1}(x)$. [3]

3. [Maximum mark: 6]

The number of registered users of a new social network is modelled by

$$N(t) = 8000 e^{kt}, \quad k > 0,$$

where t is the number of years after launch. After 3 years there are 24 000 registered users.

- (a) Find the value of k . Give your answer in the form $\frac{1}{3} \ln a$, where $a \in \mathbb{Z}^+$. [2]
- (b) Find the number of users 7 years after launch, correct to the nearest integer. [2]
- (c) Find the time, in years, at which the number of users first reaches 72 000. [2]

.....

4. [Maximum mark: 6]

Sofia invests \$2500 in an account that pays interest of 4 % per annum, compounded annually. No further deposits or withdrawals are made.

(a) Show that the value of the account at the end of the first year is \$2600. [1]

(b) Find the value of the account at the end of 10 years, correct to the nearest cent. [2]

(c) Find the smallest number of full years required for the value of the account to first reach \$5000. [3]

.....
.....
.....
.....
.....
.....
.....
.....
.....
.....

5. [Maximum mark: 5]

The Richter magnitude M of an earthquake is related to its intensity I by

$$M = \log_{10}\left(\frac{I}{I_0}\right),$$

where I_0 is a fixed reference intensity.

- (a) Earthquake A has magnitude $M = 5.4$. Find the value of $\frac{I}{I_0}$ for earthquake A .
Give your answer in the form $a \times 10^k$, where $1 \leq a < 10$ and $k \in \mathbb{Z}$. [2]
- (b) Earthquake B is 500 times more intense than earthquake A . Find the magnitude of earthquake B , correct to three significant figures. [3]

6. [Maximum mark: 4]

The intensity of light at depth d metres below the surface of a lake is modelled by

$$L(d) = L_0 e^{-0.32 d}$$

where L_0 is the intensity at the surface.

- (a) Find the percentage of the surface intensity that remains at a depth of 4 metres.
[2]
- (b) Find the depth, in metres, at which the intensity has fallen to half of its surface value. Give your answer correct to three significant figures. [2]

.....
.....
.....
.....
.....
.....
.....
.....
.....
.....
.....